

Tax Incidence, Efficiency Costs, and Optimal Commodity Taxation

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Tax Incidence

- Tax incidence is the study of the effects of tax policies on prices and the welfare of individuals
- What happens to market prices when a tax is introduced or changed?
- Example: what happens when impose \$1 per pack tax on cigarettes?
- Effect on price $\Rightarrow$ distributional effects on smokers, profits of producers, shareholders, farmers, etc.
- This is positive analysis: typically the first step in policy evaluation; it is an input to later thinking about what policy maximizes social welfare.

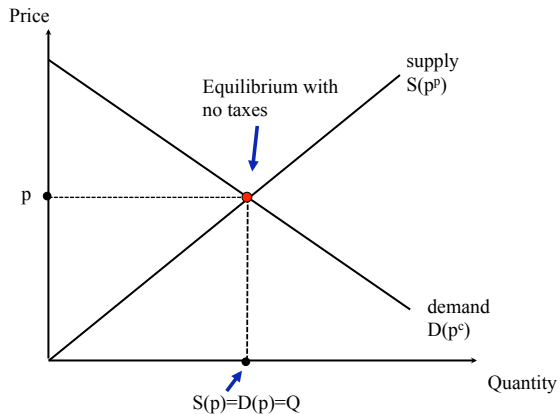
Tax Incidence

- Tax incidence is not an accounting exercise but an analytical characterization of changes in economic equilibria when taxes are changed.
- Taxes can be shifted: taxes affect directly the prices of goods, which affect quantities because of behavioral responses, which affect indirectly the price of other goods.
- If prices were constant economic incidence would be the same as legislative incidence
- Example: Does taxing corporations end up hurting workers?

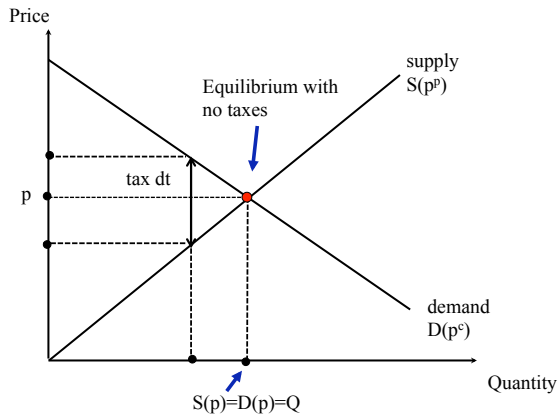
Partial Equilibrium Tax Incidence

- Partial Equilibrium Model: Simple model goes a long way to showing main results.
- Government levies an excise tax on good x
- Excise means it is levied on a quantity (gallon, pack, ton, ...). Typically fixed in nominal terms (e.g., \$1 per pack)
- Ad-valorem tax is a fraction of prices (e.g., 5% sales tax)
- Let p denote the pretax price of x (producer price)
- Let $q = p + t$ denote the tax-inclusive price of x (consumer price)

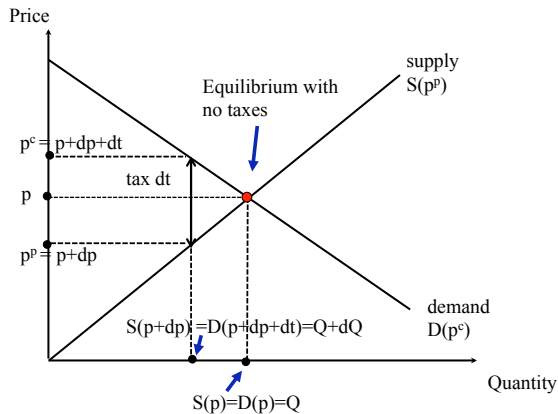
Partial Equilibrium Tax Incidence



Partial Equilibrium Tax Incidence (continued)



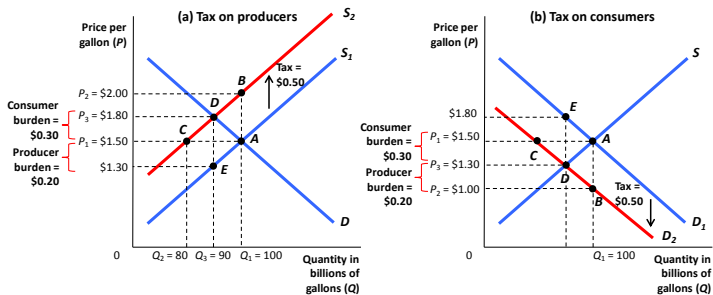
Partial Equilibrium Tax Incidence (continued)



Partial Equilibrium Tax Incidence

19.1

The Statutory Burden of a Tax Does Not Describe Who Really Bears the Tax, and Is Irrelevant to the Tax Burden



Computing Burden Shares

- For either side of the market: $[\text{burden share}] = [\text{dollars lost per unit}] / [\text{increase in tax per unit}]$
- **Consumers:** $[\text{consumers' dollars lost per unit}] = [\text{increase in tax per unit paid by consumers}] + [\text{increase in price paid per unit by consumers to producers}]$
- **Producers:** $[\text{producers' dollars lost per unit}] = [\text{increase in tax per unit paid by producers}] - [\text{increase in price paid per unit by consumers to producers}]$
- Ex. from part (b) in previous slide:
 - Consumer burden share $= [\$0.50 + (\$1.30 - \$1.50)] / [\$0.50] = 60\%$
 - Producer burden share $= [\$0.00 - (\$1.30 - \$1.50)] / [\$0.50] = 40\%$

Tax Incidence

- Demand for good x is $D(q)$ decreases with $q = p + t$
- Supply for good x is $S(p)$ increases with p
- Equilibrium condition: $Q = S(p) = D(p + t)$
- Taxes on consumers and producers have the same incidence

Tax Incidence

- Demand for good x is $D(q)$ decreases with $q = p + t$
- Supply for good x is $S(p)$ increases with p
- Equilibrium condition: $Q = S(p) = D(p + t)$
- Start from $t = 0$ and $S(p) = D(p)$. We want to characterize dp/dt : effect of a small tax increase on price, which determines who bears effective burden of tax:
- Change dt generates change dp so that equilibrium holds:

$$\begin{aligned} S(p + dp) &= D(p + dp + dt) \\ \Rightarrow S(p) + S'(p)dp &= D(p) + D'(p)(dp + dt) \\ \Rightarrow S'(p)dp &= D'(p)(dp + dt) \\ \Rightarrow \frac{dp}{dt} &= \frac{D'(p)}{S'(p) - D'(p)} \end{aligned}$$

Tax Incidence

- Useful to use elasticities in economics because elasticities are unit free
- **Elasticity:** percentage change in quantity when price changes by one percent.
- Demand elasticity: $\varepsilon_D = \frac{q}{D} \frac{dD}{dq} = \frac{qD'(q)}{D(q)} < 0$ (consumer faces price $q = p + t$).
- Supply elasticity: $\varepsilon_S = \frac{p}{S} \frac{dS}{dp} = \frac{pS'(p)}{S(p)} > 0$.
- Relation between price response and elasticities: $\frac{dp}{dt} = \frac{D'(p)}{S'(p) - D'(p)} = \frac{\varepsilon_D}{\varepsilon_S - \varepsilon_D}$.
- Bounds: $-1 \leq \frac{dp}{dt} \leq 0$ and $0 \leq \frac{dq}{dt} = 1 + \frac{dp}{dt} \leq 1$.

Tax Incidence

$$\frac{dp}{dt} = \frac{\varepsilon_D}{\varepsilon_S - \varepsilon_D}$$

When do consumers bear the entire burden of the tax? ($dp/dt = 0$ and $dq/dt = 1$)

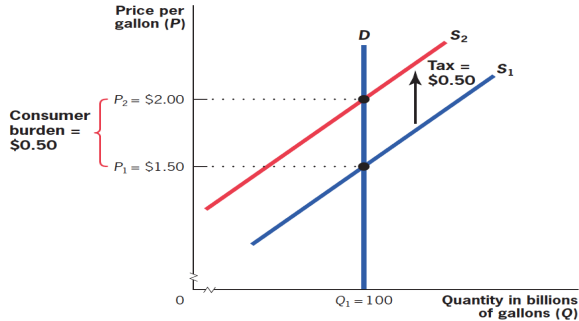
- 1 $\varepsilon_D = 0$ [inelastic demand]
example: short-run demand for gasoline inelastic (need to drive to work)
- 2 $\varepsilon_S = \infty$ [perfectly elastic supply]
example: perfectly competitive industry

When do producers bear the entire burden of the tax? ($dp/dt = -1$ and $dq/dt = 0$)

- 1 $\varepsilon_S = 0$ [inelastic supply]
example: fixed quantity supplied
- 2 $\varepsilon_D = -\infty$ [perfectly elastic demand]
example: there is a close substitute, and demand shifts to this substitute if price changes.

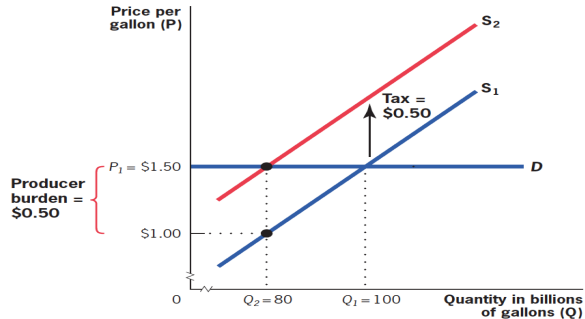
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Perfectly Inelastic Demand



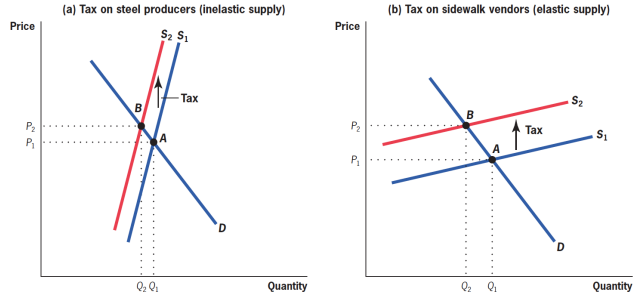
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Perfectly Elastic Demand



19.1

Supply Elasticities

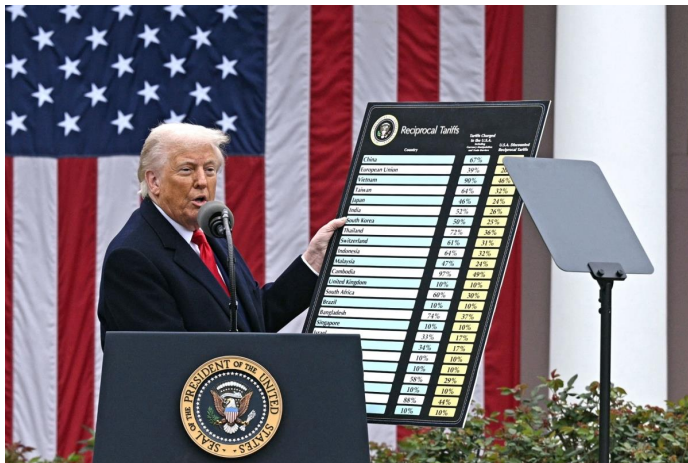


Tax Incidence: Key Results

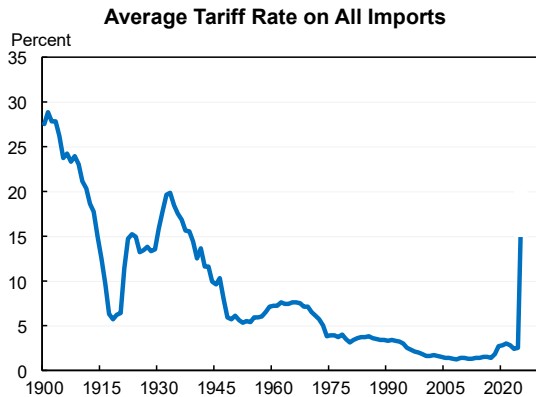
- ① Statutory incidence not equal to economic incidence
- ② Equilibrium is independent of who nominally pays the tax
- ③ More inelastic factor bears more of the tax

These are robust conclusions that hold with more complicated models

Tariffs are Commodity Taxes at the Border



They Were Mostly Tax of the Past



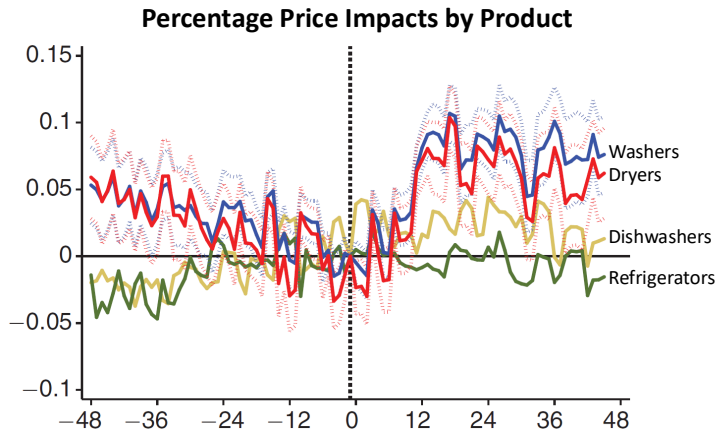
Source: The Budget Lab

Example: 2018 Tariff on Laundry Machines from LG and Samsung

- February 7, 2018: 20% tariff on all imported washing machines
- What did it do to consumer prices? Let's go to the data.



Consumers Bore Full Incidence of the Tariff



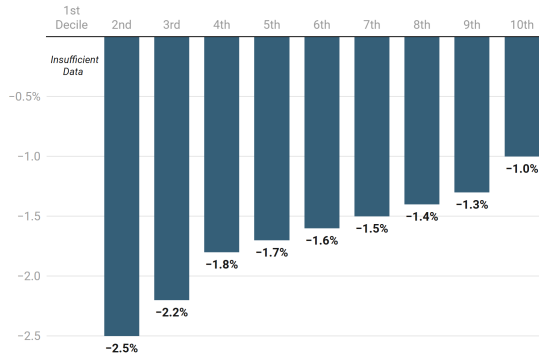
Source: Flaaen, Hortacsu, and Tintelnot, *American Economic Review*, 2020

Standard Estimates Are that Tariffs Are Regressive

Short-Term Impact of Current Tariffs on Spending Power

Through June 1

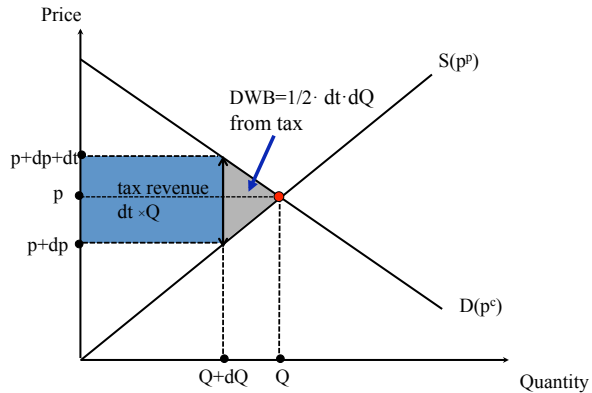
Percentage points of disposable income by household income decile



Source: The Budget Lab

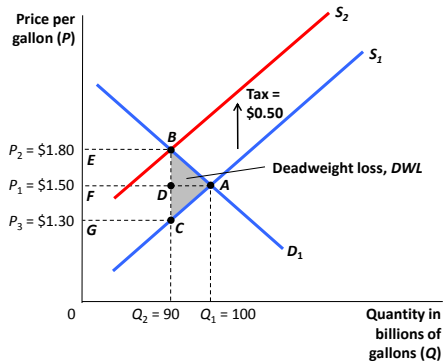
Efficiency Costs of Taxation

- Deadweight burden (also called excess burden) of taxation is defined as the welfare loss (measured in dollars) created by a tax over and above the tax revenue generated by the tax.
- In the simple supply and demand diagram, welfare is measured by the sum of the consumer surplus and producer surplus.
- The welfare loss of taxation is measured as change in consumer+producer surplus minus tax collected: it is the triangle on the figure.
- The inefficiency of any tax is determined by the extent to which consumers and producers change their behavior to avoid the tax; deadweight loss is caused by individuals and firms forgoing transactions in order to avoid taxation.
- If there is no change in the number of transactions (i.e., the quantities consumed), the tax has no efficiency costs.



20.1

Taxation and Economic Efficiency: Graphical Approach



Efficiency Costs of Taxation

- Deadweight burden (or deadweight loss) of small tax dt (starting from zero tax) is measured by the **Harberger Triangle**:

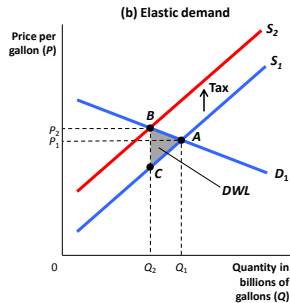
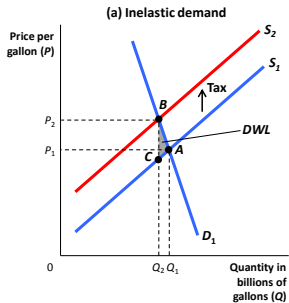
$$DWB = \frac{1}{2} (-dQ) \cdot dt = -\frac{1}{2} S'(p) \cdot dp \cdot dt = -\frac{1}{2} \frac{pS'(p)}{S(p)} \cdot \frac{Q}{p} \cdot dp \cdot dt$$

- Recall that $Q = S(p)$ and hence $dQ = S'(p)dp$.
- Recall that $dp/dt = \varepsilon_D/(\varepsilon_S - \varepsilon_D)$, hence:

$$DWB = -\frac{1}{2} \cdot \frac{\varepsilon_S \cdot \varepsilon_D}{\varepsilon_S - \varepsilon_D} \cdot \frac{Q}{p} (dt)^2$$

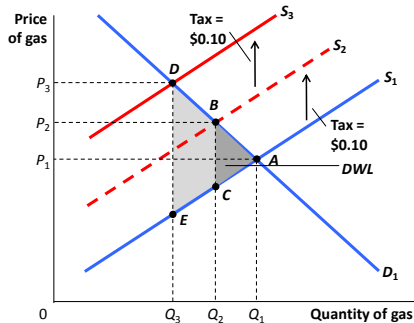
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Elasticities Determine Tax Inefficiency



20.1

Marginal DWL Rises with Tax Rate



$$DWB = -\frac{1}{2} \cdot \frac{\varepsilon_S \cdot \varepsilon_D}{\varepsilon_S - \varepsilon_D} \cdot \frac{Q}{p} (dt)^2$$

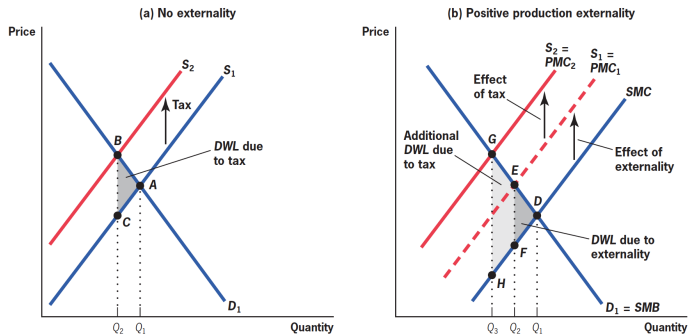
- ① DWB increases with the absolute size of elasticities $\varepsilon_S > 0$ and $-\varepsilon_D > 0$
 - ① More efficient to tax relatively inelastic goods
 - ② DWB increases with the square of the tax rate t : small taxes have relatively small efficiency costs, large taxes have relatively large efficiency costs
 - ③ More efficient to spread taxes across all goods to keep each tax rate low
 - ④ Better to fund large one time govt expense (such as a war) with debt and repay slowly afterwards than have very high taxes only during war
- ② Pre-existing distortions (such as an existing tax or externality) makes the cost of taxation higher: move from the triangle to trapezoid

Efficiency Costs of Taxation

20.1

CHAPTER 20 ■ TAX INEFFICIENCIES AND THEIR IMPLICATIONS FOR OPTIMAL TAXATION

A Tax System's Efficiency Is Affected by a Market's Preexisting Distortions



Deadweight loss is determined by quantity changes, not price changes.

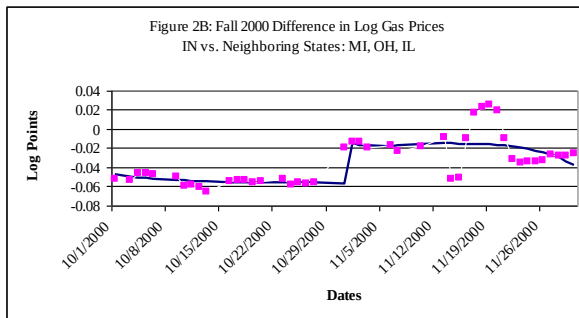
- A. True
- B. False
- C. Uncertain

Tax Incidence: Empirical Application

- Doyle and Sampatharank (2008) study the Gas Tax Holidays in Indiana (IN) and Illinois (IL).
- These gas taxes are taxes on producers (statutory incidence is on producers since gas stations pay the government).
- Are gas tax cuts passed through to consumers? Or do producers pocket the tax cut and leave consumer price unchanged?
- Study this question using state-level gas tax reforms:
 - Gas prices spike above \$2.00 in 2000
 - IN suspends 5% gas tax on July 1. Reinstated on Oct 30.
 - IL suspends 5% gas tax on July 1. Reinstated on Dec 31.

Tax Incidence: Empirical Application

- Empirical approach in paper: difference-in-difference (DD), compare treated states (e.g., IN) with neighboring states (MI, OH, MO, IA, WI) before and after tax change (e.g., expiration of the IN gas tax holiday)
- Graphical evidence is most transparent. Findings:
 - ① 10 cent increase in gas tax $\Rightarrow$ 7 cent increase in price paid by consumers
 - ② So consumers bear 70% of the burden of the gas tax: $([\$0.00] + [\$0.07]) / [\$0.10] = 70\%$ (applying formula from “Computing Burden Shares” slide)
- (Note: Consumers bearing 70% of gas tax burden means that consumers get 70% of the benefit of a gas tax cut.)



Source: Doyle and Samphantharak 2008.

19.4

EVIDENCE: The Incidence of Excise Taxation

- Excises tax on cigarettes varies widely across the United States.
 - Low of \$0.025/pack per pack in VA.
 - High of \$1.51/pack in CT and MA.
 - Since 1990, NJ increased its tax rate nearly sixfold.
 - Arizona has increased its tax nearly eightfold.
- Many studies examine how taxes affect prices.
- These studies uniformly conclude that the price of cigarettes rises by the full amount of the excise tax.

General Equilibrium Tax Incidence

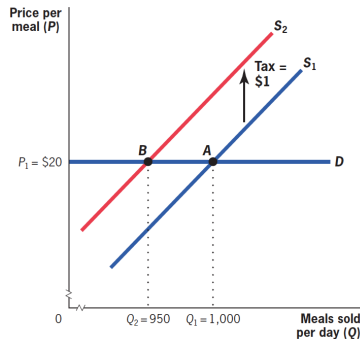
- Examples so far have focused on **partial equilibrium** incidence which considers impact of a tax on one market in isolation.
- **General equilibrium** models consider the effects on related markets of a tax imposed on one market.
- E.g., imposition of a tax on cars may reduce demand for steel $\Rightarrow$ additional effects on prices in equilibrium beyond car market.

General Equilibrium Tax Incidence: Example: Soda Tax in Berkeley

- Consider the market for Soda in Berkeley.
- Berkeley imposes a Soda tax since 2015: \$.01 per ounce ($=\$0.12/\text{can}$).
- Goal was to reduce soda consumption for better health (people overdrink); see Allcott et al. '18 for merits of soda tax.
- Here the narrower question is: Who bears the incidence?
- If soda demand is inelastic, then consumers bear the burden.
- Demand for Soda in Berkeley is likely to be elastic: if price of Soda in Berkeley goes up, you consume less Soda (intention of the tax) or buy Soda in Oakland.
- Consider the extreme case of perfectly elastic demand.

19.3

Effects of a Restaurant Tax: A General Equilibrium Example



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General Equilibrium Tax Incidence: Example: Soda Tax

- If Soda demand perfectly elastic then:
 - ① Soda sellers (supermarkets, restaurants) bear the full burden of the tax.
 - ② But Soda sellers are not self-contained entities
- Companies are just a technology for combining capital and labor to produce an output.
- Capital: land, physical inputs like building, kitchen equipment, etc.
- Labor: cashier staff, cooks, waitstaff, etc.
- ③ Ultimately, these two factors (capital or labor) must bear the loss in profits due to the tax [if consumer demand is perfectly elastic]

General Equilibrium Tax Incidence: Example: Soda Tax

- Incidence is “shifted backward” to capital and labor.
- Assume that labor supply is perfectly elastic because Berkeley restaurant/supermarket workers can always go and work in Oakland if they get paid less in Berkeley
- Capital, in contrast, is perfectly inelastic in short-run: you cannot pick up the shop and move it in the short run.
- In short run, capital bears tax because it is completely inelastic $\Rightarrow$ Soda business owners lose (not consumers or workers)
- In the longer-run, the supply of capital is also likely to be highly elastic: Investors can close or sell the shop, take their money, and invest it elsewhere.

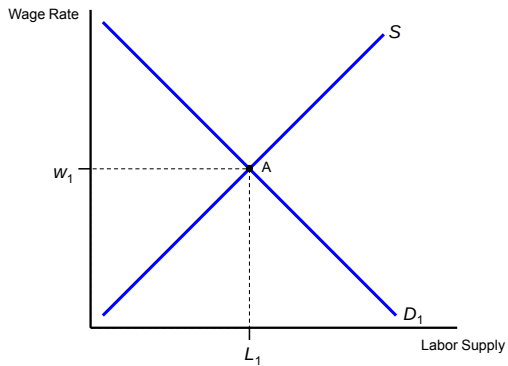
General Equilibrium Tax Incidence: Long-run effects

- If both labor and capital are highly elastic in the long run, who bears the tax?
- The one additional inelastic factor is land.
- The supply is clearly fixed.
- When both labor and capital can avoid the tax, the only way Soda sellers will remain in Berkeley is if they pay a lower rent on their land.
- $\Rightarrow$ Soda tax ends up hurting Berkeley landowners in general equilibrium [if Soda demand, labor and capital are fully elastic]
- This is of course an idealized example, in practice, demand, labor, and capital are not fully elastic

Mandated Benefits

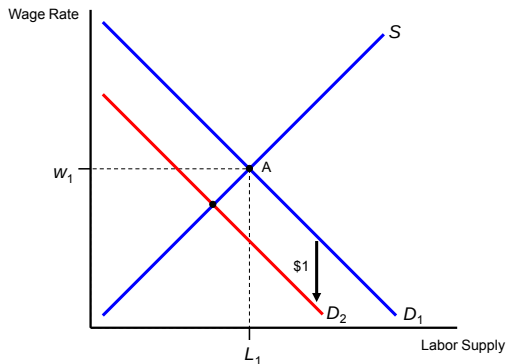
- Now consider incidence of a mandated benefit instead of a tax
- Examples:
 - ① requirement that employers pay for healthcare (employers with 50+ employees are required to do that under Obamacare law or pay a fine),
 - ② workers compensation benefits [for injuries on the job]
- Affects firms like a tax
- But effect of mandated benefits on equilibrium wages and employment differ from a tax (Summers 1989) because workers value the mandated benefit
- Suppose workers value \$1 of mandated benefit at \$ $\alpha \geq 0$
- Could have $\alpha < 1$ if benefit not as valuable as cash
- Could have $\alpha > 1$ if benefit more valuable than cash (e.g., can't buy health insurance on individual market)
- If $\alpha = 1$ then no change in employment

Figure 1: Mandated Benefit



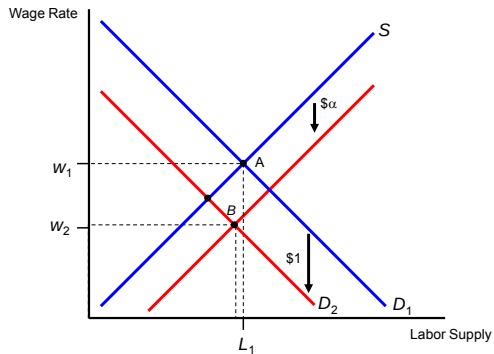
Source: Raj Chetty's Undergraduate Lecture Note

Figure 1: Mandated Benefit



Source: Raj Chetty's Undergraduate Lecture Note

Figure 1: Mandated Benefit



Source: Raj Chetty's Undergraduate Lecture Note

Tax Salience: A New Theory

- Traditional model assumes individuals are fully aware of taxes they pay.
- In practice, many taxes (e.g., sales tax) are not salient to consumers.
- Examples:
 - Do you know your marginal income tax rate?
 - Do you consider it when choosing a job?
 - Do you know the sales tax added at checkout?
- Chetty, Looney, and Kroft (AER '09) study salience and develop a model with inattentive consumers.

Tax Salience: A New Theory

- Chetty, Looney, and Kroft (AER '09) use two empirical strategies to test salience:
- Randomized field experiment:
 - Treatment stores display price tags showing sales tax and total price for a subset of products.
 - Compare treated vs control products before and after the change (difference-in-difference); use control stores as placebo.
- Policy experiment:
 - Exploit variation in beer excise taxes (salient, included in posted price) vs sales taxes (not included in posted price).
 - Compare consumer responses to excise vs sales tax changes.



Source: Chetty, Looney, Kroft (2009)

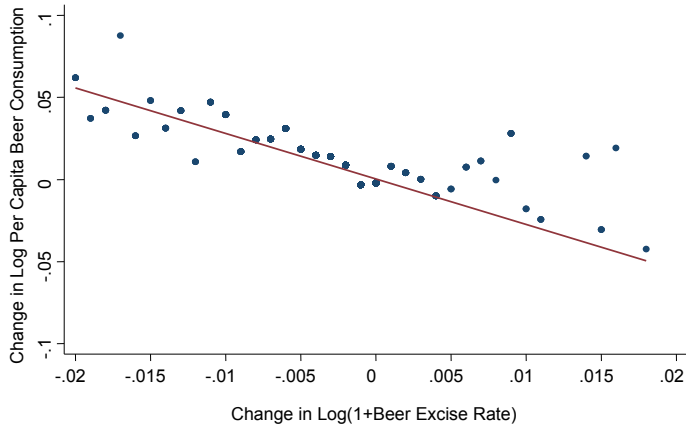
Effect of Posting Tax-Inclusive Prices: Mean Quantity Sold

TREATMENT STORE			
Period	<u>Control Categories</u>	<u>Treated Categories</u>	<u>Difference</u>
Baseline	26.48 (0.22)	25.17 (0.37)	-1.31 (0.43)
Experiment	27.32 (0.87)	23.87 (1.02)	-3.45 (0.64)
Difference over time	0.84 (0.75)	-1.30 (0.92)	DD_{TS} = -2.14 (0.64)
CONTROL STORES			
Period	<u>Control Categories</u>	<u>Treated Categories</u>	<u>Difference</u>
Baseline	30.57 (0.24)	27.94 (0.30)	-2.63 (0.32)
Experiment	30.76 (0.72)	28.19 (1.06)	-2.57 (1.09)
Difference over time	0.19 (0.64)	0.25 (0.92)	DD_{CS} = 0.06 (0.90)
DDD Estimate			-2.20 (0.58)

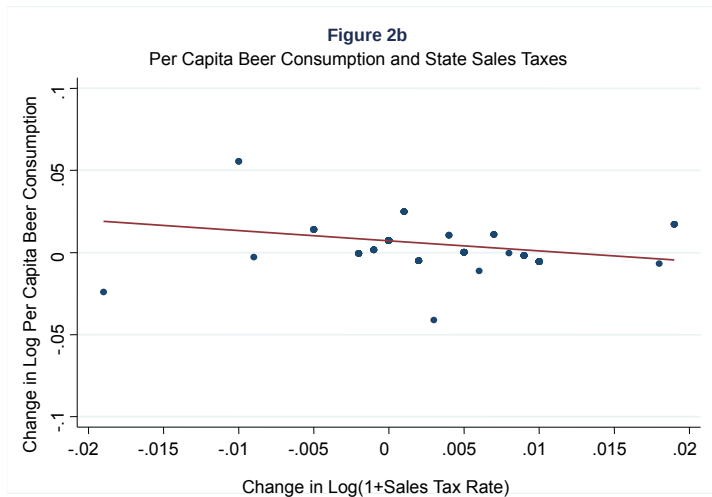
Source: Chetty, Looney, Kroft (2009)

Figure 2a

Per Capita Beer Consumption and State Beer Excise Taxes



Source: Chetty, Looney, Kroft (2009)



Source: Chetty, Looney, Kroft (2009)

Effect of Excise and Sales Taxes on Beer Consumption

Dependent Variable: Change in Log(per capita beer consumption)				
	Baseline	Bus Cyc, Alc Regs.	3-Year Diffs	Food Exempt
	(1)	(2)	(3)	(4)
$\Delta \text{Log}(1 + \text{Excise Tax Rate})$	-0.87	-0.89	-1.11	-0.91
	(0.17)***	(0.17)***	(0.46)**	(0.22)***
$\Delta \text{Log}(1 + \text{Sales Tax Rate})$	-0.20	-0.02	-0.00	-0.14
	(0.30)	(0.30)	(0.32)	(0.30)
Business Cycle Controls		x	x	x
Alcohol Regulation Controls		x	x	x
Year Fixed Effects	x	x	x	x
F-Test for Equality of Coeffs.	0.05	0.01	0.05	0.04
Sample Size	1,607	1,487	1,389	937

Source: Chetty, Looney, Kroft (2009)

Note: Estimates imply $\theta_{\tau} \approx 0.06$

Key Empirical Result: Salience Matters

- ① Posting sales taxes reduces demand for those goods
⇒ Breaks tax incidence result #2: equilibrium depends on who nominally pays the tax
- ② Beer consumption is elastic to excise tax rate (built in posted price) but not to the sales tax rate (not built in the posted price)
⇒ If tax is not salient to consumers, they are less elastic, and hence more likely to bear the tax burden

A number of recent empirical studies show that individuals are not fully informed and fully rational and this has large consequences for policy

Should All Goods Be Taxed at the Same Rate?

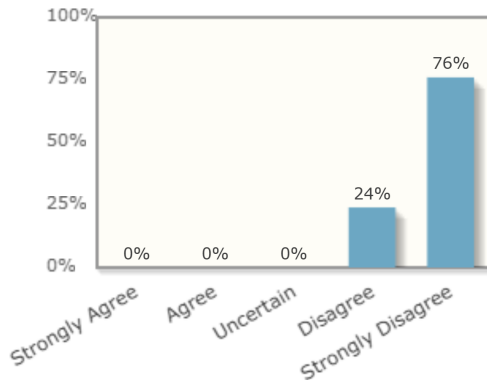
- U.S. governments tax income, sales (commodities), and property
- We will discuss property taxes later
- Sales taxes are mostly flat, leaving redistribution to the income tax-and-transfer system
- But we tax imports higher than domestic goods
- Most sales taxes exempt groceries, and we could also tax Rolex watches
- Should we tax all commodities at the same rate?

Should All Goods Be Taxed at the Same Rate?

- Atkinson-Stiglitz (1976) provide the key way to think about this question. Laroque (2005) simplified it.
- Key lesson: If the reason that rich people buy Rolexes is that they have a lot of money, then there is no benefit to taxing Rolexes differently from groceries
- Instead, all redistribution should be done through the income tax-and-transfer system
- Same logic applies to tariffs
- Simple model (e.g., zero producer incidence), but it is the key reason that economists focus on the income tax-and-transfer system

Economists Don't Like Tariffs

2018 poll of top economists: “Imposing new US tariffs on steel and aluminum will improve Americans’ welfare.”



Source: University of Chicago Booth School of Business, “Steel and Aluminum Tariffs”, 2018

Economists Don't Like Tariffs



- Austan Goolsbee, Pres. Obama's chief economist
- Explains his poll response: "SMACK. SMACK [punching self in face] SMACK. SMACK."



- Greg Mankiw, President Bush's chief economist
- "The Wall Street Journal rightly calls the new Trump tariffs 'the dumbest trade war in history'."

Source: Chicago Booth "Steel and Aluminum Tariffs", 2018. Greg Mankiw's Blog, 2025.

Model setup

- Consumers i choose l_i hours to work and spend their after-income-tax income on either groceries g_i or Rolex watches r_i :

$$\max_{g_i, r_i, l_i} U^i(v(g_i, r_i), l_i)$$

$$\text{s.t. } g_i + (p + t)r_i = w_i l_i - T(w_i l_i)$$

- Consumers differ by wages w_i and by U^i but share same subutility across groceries and Rolexes $v(g_i, r_i)$
- Government has a nonlinear labor income tax $T(z)$ and an ad-valorem tax on Rolexes t

Indirect Utility

- Let $V(z, p + t) = \max_{g, r} v(g, r)$ such that $g + (p + t)r \leq z - T(z)$ be the indirect utility of consumption vector $\{g, r\}$
- Indirect utility $V(z, p + t)$ is common to all individuals
- Everyone who earns z spends it in exactly the same way, because $v(g, r)$ is the same for all individuals
- Rich people buy Rolexes because they have more money, not because they have a stronger preference for Rolexes

Strategy: Proof by Contradiction

- Take an initial tax system $(T(\cdot), t)$ with a Rolex tax $t \neq 0$. Let $g(t)$ and $r(t)$ be consumer choice under that initial tax system.
- Suppose that this initial tax system is optimal.
- Show that society can do better without the Rolex tax
- Show that the government can provide individuals the same menu of labor disutility and consumption utility more efficiently (i.e., collects more taxes) with $t = 0$

Step 1/3: Define an Alternative Tax System

- Define a specific alternative tax system: Replace $(T(.), t)$ with $(\bar{T}(.), t = 0)$ where $\bar{T}(z)$ is chosen such that indirect utility is held constant at every earnings choice z :
$$V(z - T(z), p + t) = V(z - \bar{T}(z), p)$$
 - Concretely, suppose I currently pay \$2,000 in Rolex taxes
 - One alternative tax system would be to increase my income tax by \$2,000
 - The question here is whether the government can increase my income tax by more than \$2,000 and still leave me with the same utility as before
- Utility $U^i(v, z)$ unchanged for every individual i at every earnings (labor supply) choice z .
Utility and earnings unchanged for all. Only effect can be on the government budget.

Step 2/3: Observe That the New Consumption Bundle is Cheaper

- By definition of indirect utility, attaining $V(z - \bar{T}(z), p)$ at price p costs $z - \bar{T}(z)$, via new consumption $\bar{g}(0), \bar{r}(0)$: $z - \bar{T}(z) = g(0) + p \cdot \bar{r}(0)$
- The new consumption bundle costs less than the old consumption bundle which delivers the same consumption utility, or else would choose the old consumption bundle:
 $g(0) + p \cdot \bar{r}(0) \leq g(t) + p \cdot r(t)$

Step 3/3: Observe That the Government Collects More Taxes

- All consumers attain the same utility $V(z - \bar{T}(z), p) = V(z - T(z), p + t)$ when choosing $g(t), r(t)$ under old tax system: $z - T(z) = g(t) + (p + t) \cdot r(t)$
- But government collects more taxes under new tax system
 - $z - \bar{T}(z) = p \cdot \bar{c}(0) \leq p \cdot c(t) = z - T(z) - t \cdot c(t)$
 - $\bar{T}(z) \geq T(z) + t \cdot c(t)$
- Hence, the initial tax system cannot be optimal.
- What happened here? The government was able to tax away social savings from not distorting consumption.

- Baseline model says to do redistribution through the income tax-and-transfer system, not through differentiated commodity taxes
- We now turn to income taxes and transfers
- Still clear scope for differential commodity taxes when there are externalities (e.g., pollution)
- Scope for differential commodity taxes with producer incidence?

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